PROJECT DESIGN PHASE

Project Proposal (Proposed Solution)

Date	15 March 2024
Team ID	
Project Title	Global Fuel Prices and Taxation Trends: A Comprehensive Analysis
Maximum Marks	3 Marks

Project Proposal

This proposal defines a Tableau-based analytical solution for comparing global fuel prices and examining how taxation, subsidy levels, crude-oil prices, region and income level relate to price differences. The project uses descriptive analysis to make a multi-country dataset easier to explore and interpret.

Section	Project Details
Project Overview	A data-analysis and visualization project that explores petrol, diesel and LPG prices across countries from 2020–2026, alongside Brent crude oil prices and tax percentages.
Objective	To identify and communicate patterns in fuel prices across countries and time, and compare those patterns with taxation, subsidy level, region and income level using Tableau visualizations.
Scope	Includes dataset review, preparation, country and time-based comparisons, fuel-type analysis, tax/subsidy analysis, Brent crude comparison, an interactive Tableau dashboard and a Tableau Story. It is descriptive and does not claim that observed relationships prove causation.
Problem Statement	Fuel-price information across countries, dates and fuel types can be difficult to compare in raw tabular form. Relevant factors such as taxation, subsidy level and Brent crude prices need to be reviewed together to make trends easier to understand.
Impact	A consolidated visual analysis can help students, researchers and interested viewers compare countries, notice trends and interpret differences without manually scanning thousands of records.
Proposed Solution	Prepare the global fuel-price CSV for analysis, build Tableau worksheets for key questions, combine selected views in a dashboard, and present findings sequentially in a Tableau Story. A Flask/Bootstrap project website can provide a supporting project overview and links.
Approach	1. Inspect the dataset and fields. 2. Check data quality and types. 3. Create comparative and trend visualizations. 4. Assemble the dashboard and Story. 5. Review labels, units, filters and findings. 6. Publish and document the outputs.
Key Features	Country-level average petrol prices; global fuel-price trends; petrol/diesel/LPG comparison; tax percentage and subsidy analysis; views by region/income level; Brent crude trend comparison; interactive dashboard and explanatory Story.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Computer for data preparation, Tableau authoring and web development.	Existing laptop/PC; no dedicated GPU or server is required for this dataset.
Computing Resources	Run Tableau, browser and development tools.	Local development environment; exact CPU/core count not specified.
Memory	Support data loading and visualization work.	Use available system RAM; no fixed minimum was documented.
Storage	Store CSV, workbook, website files and project documents.	Local SSD and GitHub repository for versioned project files.
Software	Tools for visualization and supporting website.	Tableau Public/Desktop, Python, Flask, HTML, CSS, Bootstrap, web browser.
Frameworks	Web presentation framework.	Flask and Bootstrap for the supporting project site.
Libraries	Libraries used in the analysis or web app.	Only list libraries actually used in the repository; do not assume unconfirmed packages.
Development Environment	Editing, testing and source control.	Local code editor and Git/GitHub; exact editor version not specified.
Data	Structured fuel-price dataset.	global_fuel_prices_Dataset.csv; 27,468 records, 84 countries, 10 columns; date range 2020–2026.

Assumption note: Specific hardware specifications and library versions were not provided, so they are not invented in this proposal.